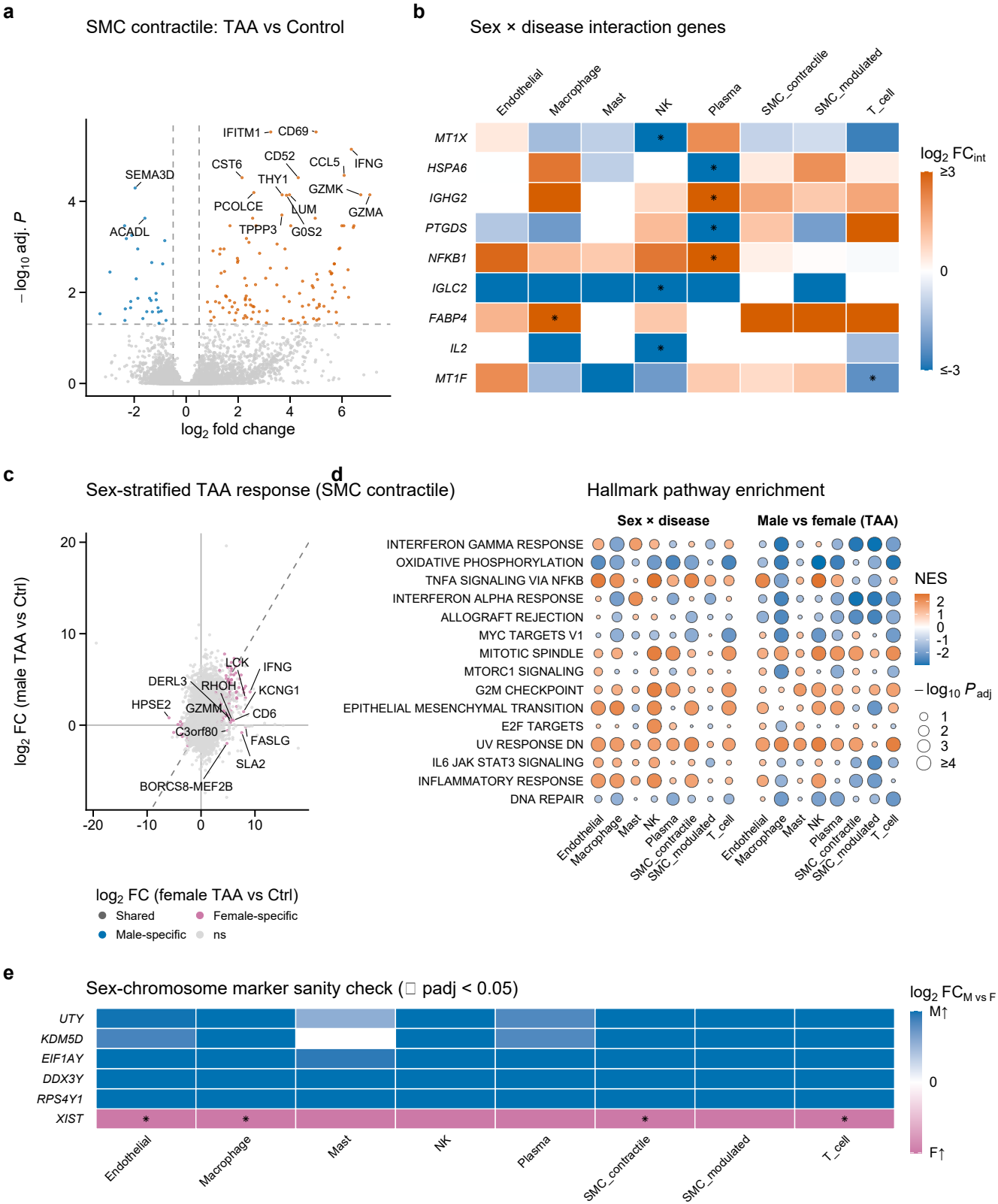

a
SMC contractile: TAA vs Control
IFITM1
CD69
CD52
CCL5
IFNG
CST6
SEMA3D
THY1
GZMK
PCOLCE
LUM
GZMA
ACADL
TPPP3
G0S2
− log10 adj. P
log2 fold change
b
Sex × disease interaction genes
Endothelial
Macrophage
Mast
NK
Plasma
SMC_contractile
SMC_modulated
T_cell
MT1X
HSPA6
IGHG2
PTGDS
NFKB1
IGLC2
FABP4
IL2
MT1F
log2 FCint
≥3
0
≤-3
c
Sex-stratified TAA response (SMC contractile)
log2 FC (male TAA vs Ctrl)
LCK
IFNG
DERL3
RHOH
KCNG1
HPSE2
GZMM
CD6
C3orf80
FASLG
SLA2
BORCS8-MEF2B
log2 FC (female TAA vs Ctrl)
Shared
Female-specific
Male-specific
ns
d
Hallmark pathway enrichment
Sex × disease
Male vs female (TAA)
INTERFERON GAMMA RESPONSE
OXIDATIVE PHOSPHORYLATION
TNFA SIGNALING VIA NFKB
INTERFERON ALPHA RESPONSE
ALLOGRAFT REJECTION
MYC TARGETS V1
MITOTIC SPINDLE
MTORC1 SIGNALING
G2M CHECKPOINT
EPITHELIAL MESENCHYMAL TRANSITION
E2F TARGETS
UV RESPONSE DN
IL6 JAK STAT3 SIGNALING
INFLAMMATORY RESPONSE
DNA REPAIR
NES
− log10 Padj
e
Sex-chromosome marker sanity check (□ padj < 0.05)
UTY
KDM5D
EIF1AY
DDX3Y
RPS4Y1
XIST
log2 FCM vs F
M↑
F↑